

User Study – ML2++ (Textual DSL)

Instructions: Fill Section A once. For Section B, answer for *each* use case you worked on. If you did multiple use cases, **duplicate Section B** per use case.

Participant ID: User3 Use case(s): ☐ U1 River Flow ☐ U2 Smart-Home Energy
☐ U3 Solar Energy

Section A – Background (answer once)

This section helps us understand your experience level so we can analyze results fairly.

A1. Educational level (highest completed)

Tell us the highest level of education you have finished.

☐ High School ☐ Vocational/Technical ☒ Bachelor's ☐ Master's ☐ Doctorate ☐ Other: _____

A2. Primary role (pick the closest)

Choose the option that best matches your current job or main activity. If none fits, select “Other”.

☐ IoT/CPS Developer ☐ MDE Practitioner ☒ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: _____

A3. Self-rated expertise (tick one per row)

For each area, select your skill level. “IoT/CPS” = connected devices/systems. “MDE” = using models to guide or generate code. “ML/Data Science” = working with data and ML.

	Beginner	Intermediate	Advanced	Expert
IoT/CPS development	☒	☐	☐	☐
Model-Driven Engineering (MDE)	☒	☐	☐	☐
Machine Learning / Data Science	☒	☐	☐	☐

A4. Prior experience (used DSLs/modeling tools/ML frameworks before?)

Have you used similar tools (e.g., modeling editors, DSLs, ML libraries/frameworks)?

☐ Yes ☒ No

Section B – Task-Specific (duplicate this page *per use case*)

This B-section is for: ☐ U1 River Flow ☒ U2 Smart-Home Energy ☐ U3 Solar Energy

If you did more than one use case, copy this whole Section B for each one and fill it separately.

B1. Time & Effort

Please record your times honestly. If unsure, give your best estimate. “Extra learning time” = time reading docs/tutorials to figure things out.

B1. Time spent (minutes)

Minutes you spent defining/editing the DSL, generating code, reviewing outputs, and any extra learning time.

	ML2++ (Textual DSL)
Defining DSL + generate + review	400min
Extra learning time	60min

B2. Statements written (DSL) (exclude blanks/comments)

Approximate number of DSL lines/statements you actually typed or edited.

	ML2++ (Textual DSL)
ML2++ DSL statements	100min

B2. Task Completion & Ease of Use

Tell us if you finished the task and how easy/hard each step felt. Then write how many minutes you spent per step.

B3. Were you able to finish the use case?

“None” = could not complete; “Partial” = finished some parts; “Complete” = finished everything.

☐ None ☐ Partial ☒ Complete

B4. Ease by workflow step (with per-step time)

Rate difficulty (1=very easy, 5=very difficult) and write minutes spent. “Model configuration” = choosing algorithm and hyperparameters. “Model evaluation” = checking results with plots/metrics.

Step	1	2	3	4	5	Time
Whole workflow (end-to-end)	☐	☐	☐	☒	☒	600min
Data preprocessing	☐	☐	☐	☒	☐	100min
Model configuration (algorithm, hyperparameters)	☐	☐	☐	☒	☐	200min
Model training	☐	☐	☐	☒	☐	200min
Model evaluation (metrics/plots/inspection)	☐	☐	☐	☒	☐	100min

Tip: use a simple stopwatch and pause for breaks. If unsure, estimate.

B5. Ease of ML2++ elements (1=Very easy → 5=Very difficult)

Rate how easy it was to work with each part of the language/editor.

Element	1	2	3	4
Data types & objects	☐	☒	☐	☐
Things & behaviours (statecharts)	☐	☐	☒	☐
Ports & properties	☐	☐	☒	☐
Model configuration options (algorithm/hyperparameters)	☐	☐	☐	☒
Configuration (instances/connectors)	☐	☐	☐	☒

B3. Learning & Usability

These questions focus on how quickly you learned the tool and how natural it felt.

B6. Learning curve (1=Very easy → 5=Very difficult)

How much effort did it take to learn enough to finish this use case?

☐1 ☐2 ☐3 ☒4 ☐5

B7. Intuitiveness (1=Very intuitive → 5=Not intuitive)

Did the tool feel natural/obvious to use, or did it feel confusing?

☐1 ☐2 ☐3 ☒4 ☐5

B8. Hardest part to learn (briefly)

I encountered several issues primarily related to code execution. Initially, I struggled with data preprocessing—I couldn't even load or run it properly, which blocked progress and took considerable time to diagnose (it was quite difficult to resolve, around a 4/5 on the effort scale). Later, I faced problems with the prediction plotting feature; the parameters I defined in the DSL kept causing errors (e.g., mismatched types or invalid ranges), making it frustrating and time-consuming to fix, as the error messages weren't always specific enough (effort level about 3/5). Overall, these debugging hurdles made the workflow feel clunky, but I managed to work around them by trial-and-error adjustments.

B4. Debugging & Error Handling

Tell us about problems you faced and how hard they were to fix.

B9. Error frequency (0=None, 1=Few, 2=Moderate, 3=Many)

How many problems blocked you (errors, failed runs, missing packages, wrong paths, etc.)?

☐0 ☐1 ☐2 ☒3

B10. Clarity of error messages (1=Not clear → 5=Very clear)

When something went wrong, did the messages help you understand the fix?

☐1 ☐2 ☒3 ☐4 ☐5

B11. Debugging effort (1=Easy → 5=Hard)

Overall, how hard was it to fix your issues?

☐1 ☐2 ☐3 ☒4 ☐5

B12. Debugging time (minutes) (estimate)

About how many minutes did you spend fixing problems (total for this use case)?

300min

B5. Code Quality, Maintainability, Flexibility

These questions refer to the generated code you viewed and how easy it would be to change things later.

B13. Generated code quality (1=Poor → 5=Excellent)

How good is the generated code (readable, well-structured, correct)?

☐1 ☒2 ☐3 ☐4 ☐5

B14. Maintainability (1=Very difficult → 5=Very easy)

If you had to update/change your solution later (e.g., new dataset, new horizon), how easy would it be?

☐1 ☒2 ☐3 ☐4 ☐5

B15. Flexibility / Customisation (1=None → 5=Very flexible)

How easily can you change models, hyperparameters, thresholds, or add steps without workarounds?

☐1 ☐2 ☐3 ☒4 ☐5

Section C – Overall Assessment (ML2++ only)

Your overall opinion and which features helped you most.

C1. Overall satisfaction (1=Very dissatisfied → 5=Very satisfied)

Your overall feeling about using ML2++ for this use case.

☐1 ☐2 ☒3 ☐4 ☐5

C2. Would you use ML2++ for your next similar project?

If you had a similar project tomorrow, would you choose this approach? Tell us briefly why.

☐ Yes ☒ No ☐ Undecided Why? ~~The debugging process was too challenging and time-consuming, which made the overall workflow inefficient for rapid prototyping or production use.~~

C3. Feature effectiveness (1=Ineffective → 5=Very effective)

How useful were these features for you?

Feature	1	2	3	4	5
Time-series forecasting models	☐	☐	☒	☐	☐
Automatic preprocessing & plots	☐	☐	☒	☐	☐
Training configuration options	☐	☐	☒	☐	☐
Multi-day forecasting accuracy	☐	☐	☒	☐	☐

C4. Favourite feature: The time-series forecasting models feature, as it provided a solid foundation for analyzing time-based data, even though the other parts were challenging.

What was the single most helpful feature?

C5. Least favourite feature: The automatic preprocessing & plots feature, as it often failed to handle edge cases in the data loading step, leading to cryptic errors that disrupted the workflow and required manual workarounds.
Which feature felt least helpful or got in your way?

C6. Missing features you would like: _____

What do you wish the tool had (e.g., a plot, algorithm, wizard, or shortcut)?

The automatic preprocessing & plots feature, as it often failed to handle edge cases in the data loading step, leading to cryptic errors that disrupted the workflow and required manual workarounds.

Section D – Open Feedback

Tell us anything else that would help us improve.

D1. Barriers to adoption (*what would stop you from using ML2++ in production?*)

Examples: setup, missing plots, docs, integration, performance, governance.

Poor error handling and debugging tools, plus inadequate documentation for production-scale integration.

D2. One improvement you would prioritize

If we could fix/add only one thing, what would help you the most?

Improved error handling with more descriptive messages and automated suggestions for common fixes.

Scale anchors used throughout for consistency:

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.